

Longest Subarray of Evens and Odds

Difficulty: Medium | Accuracy: 59.21% | Submissions: 40K+ | Points: 4

Given an array `arr[]`, return the maximum possible length of a subarray such that its elements are arranged alternately either as even and odd or odd and even.

Examples

Input: `arr[] = [10, 12, 14, 7, 8]`

Output: 3

Explanation: The max length of subarray is 3 and the subarray is [14, 7, 8]. Here the array starts as an even element and has odd and even elements alternately.

Input: `arr[] = [4, 6]`

Output: 1

Explanation: The array contains [4, 6]. So, we can only choose 1 element as that will be the max length subarray.

Constraints

$1 \leq \text{arr.size()} \leq 10^6$

$1 \leq \text{arr}[i] \leq 10^3$

Expected Complexities

Time Complexity: $O(n)$

Auxiliary Space: $O(1)$

Topic Tags

Arrays